NCCIA Model Paper — Cybercrime Investigation Pattern (Solved)

Model / Practice Paper — Official Pattern · 100 MCQs · Time: 100 minutes · SOLVED (with answer key & explanations)

Prepared by MR Academy for StudySphere. This is an original practice paper in the official test pattern — not an official past paper.

Computer & Information Technology

1. The full form of IP in 'IP address' is:

A) Internet Program

B) Internet Protocol ✓

C) Internal Protocol

D) Internet Provider

Answer: B — Internet Protocol. IP stands for Internet Protocol.

2. Which protocol is used to send email?

A) FTP

B) SMTP ✓

C) POP3 only

D) HTTP

Answer: B — SMTP. SMTP (Simple Mail Transfer Protocol) sends email; POP3/IMAP receive it.

3. A 'phishing' attack is:

A) A fishing competition

B) A fraudulent attempt to obtain sensitive information by posing as a trustworthy entity ✓

C) A type of firewall

D) A backup method

Answer: B — A fraudulent attempt to obtain sensitive information by posing as a trustworthy entity. Phishing tricks users into revealing passwords or financial data.

4. Which of the following is a strong password practice?

A) Using your date of birth

B) Using the same password everywhere

C) Using a long mix of letters, numbers and symbols, unique per account ✓

D) Using 'password123'

Answer: C — Using a long mix of letters, numbers and symbols, unique per account. Long, unique passphrases are the safest common practice.

5. Malware that locks files and demands payment is called:

A) Spyware

B) Ransomware ✓

C) Adware

D) Worm

Answer: B — Ransomware. Ransomware encrypts files and demands a ransom.

6. Two-factor authentication (2FA) adds security by requiring:

A) Two passwords

B) A second proof, e.g. a code sent to your phone ✓

C) Two computers

D) Two accounts

Answer: B — A second proof, e.g. a code sent to your phone. 2FA combines something you know with something you have.

7. Which port is the default for HTTPS?

A) 80

B) 21

C) 443 ✓

D) 25

Answer: C — 443. HTTPS uses port 443; HTTP uses port 80.

8. The 'S' in HTTPS mainly indicates:

A) Speed

B) Secure (encrypted) connection ✓

C) Server

D) Standard

Answer: B — Secure (encrypted) connection. HTTPS encrypts data using TLS/SSL.

9. A program that records keystrokes secretly is called a:

A) Keylogger ✓

B) Codec

C) Compiler

D) Driver

Answer: A — Keylogger. Keyloggers secretly record what a user types.

10. Which of the following best defines 'digital forensics'?

A) Deleting old files

B) The recovery and investigation of material found in digital devices ✓

C) Designing websites

D) Repairing hardware

Answer: B — The recovery and investigation of material found in digital devices. Digital forensics recovers and examines digital evidence for investigations.

11. In computing, 'hashing' is used to:

A) Compress photos

B) Produce a fixed-size fingerprint of data for integrity checks ✓

C) Send emails

D) Speed up Wi-Fi

Answer: B — Produce a fixed-size fingerprint of data for integrity checks. A hash value changes if data is altered, proving tampering.

12. Which is an example of an operating system?

A) Linux ✓

B) Oracle

C) Python

D) HTML

Answer: A — Linux. Linux is an operating system; Python is a language, HTML a markup language.

13. The device that connects two different networks and forwards data packets is a:

A) Hub

B) Router ✓

C) Monitor

D) Scanner

Answer: B — Router. A router forwards packets between networks (e.g. home network and the internet).

14. 'SQL injection' is an attack on:

A) Power supply

B) Databases through input fields ✓

C) Printers

D) Keyboards

Answer: B — Databases through input fields. SQL injection inserts malicious database queries via user input.

15. Which file extension is commonly used for a compressed archive?

A) .zip ✓

B) .mp3

C) .jpg

D) .html

Answer: A — .zip. .zip is a widely used archive/compression format.

16. CPU registers are:

A) Very slow storage

B) Small, very fast storage locations inside the CPU ✓

C) External drives

D) Network devices

Answer: B — Small, very fast storage locations inside the CPU. Registers are the fastest, smallest memory inside the CPU.

17. Which language is primarily used to style web pages?

A) HTML

B) CSS ✓

C) SQL

D) C++

Answer: B — CSS. CSS (Cascading Style Sheets) styles web pages.

18. A 'botnet' is:

A) A single robot

B) A network of infected computers controlled remotely ✓

C) A type of antivirus

D) A chat program

Answer: B — A network of infected computers controlled remotely. Botnets are hijacked devices used for attacks like DDoS.

19. DDoS attack aims to:

A) Steal passwords

B) Make a service unavailable by flooding it with traffic ✓

C) Encrypt files

D) Copy databases quietly

Answer: B — Make a service unavailable by flooding it with traffic. Distributed Denial of Service floods a target until it cannot serve users.

20. Which storage is non-volatile?

A) RAM

B) Cache

C) SSD ✓

D) Register

Answer: C — SSD. An SSD keeps data without power; RAM/cache/registers are volatile.

21. The binary number 1010 equals decimal:

A) 8

B) 9

C) 10 ✓

D) 12

Answer: C — 10. 1010■ = 8 + 2 = 10.

22. Which company created the Android operating system originally (later bought by Google)?

A) Android Inc. ✓

B) Nokia

C) Samsung

D) Microsoft

Answer: A — Android Inc.. Android Inc. created Android; Google acquired it in 2005.

23. 'VPN' stands for:

A) Virtual Private Network ✓

B) Verified Public Network

C) Virtual Program Node

D) Voice Protocol Network

Answer: A — Virtual Private Network. A VPN creates an encrypted tunnel over a public network.

24. Which of the following is a cloud storage service?

A) Google Drive ✓

B) MS Paint

C) Notepad

D) VLC

Answer: A — Google Drive. Google Drive stores files in the cloud.

25. Email spoofing means:

A) Deleting emails

B) Forging the sender address of an email ✓

C) Encrypting email

D) Blocking spam

Answer: B — Forging the sender address of an email. Spoofing fakes the 'From' address to deceive recipients.

26. The first generation of computers used:

A) Transistors

B) Vacuum tubes ✓

C) Microprocessors

D) ICs

Answer: B — Vacuum tubes. First-generation computers used vacuum tubes.

27. Which symbol is used for single-line comments in Python?

A) //

B) # ✓

C) /* */

D) -- only

Answer: B — #. Python uses # for comments.

28. A database table's unique identifier for each record is the:

A) Foreign key

B) Primary key ✓

C) Index only

D) Trigger

Answer: B — Primary key. The primary key uniquely identifies each record.

29. 'Open source' software means:

A) Free of cost always

B) Source code is available to view, use and modify under a licence ✓

C) No licence needed

D) Works only online

Answer: B — Source code is available to view, use and modify under a licence. Open source licences let users inspect and modify the code.

30. Which tool is used to capture and analyse network traffic?

A) Wireshark ✓

B) Photoshop

C) Audacity

D) Blender

Answer: A — Wireshark. Wireshark is a well-known network protocol analyser.

English — Vocabulary & Grammar

31. Choose the synonym of 'Confidential':

A) Public

B) Secret ✓

C) Simple

D) Legal

Answer: B — Secret. Confidential means secret or private.

32. Choose the antonym of 'Permit':

A) Allow

B) Forbid ✓

C) Grant

D) Let

Answer: B — Forbid. Permit means allow; forbid is the opposite.

33. He was accused of ___ into the bank's system.

A) hack

B) hacking ✓

C) hacked

D) hacks

Answer: B — hacking. Prepositions are followed by the -ing form: 'accused of hacking'.

34. The data ___ stolen last night.

A) were

B) was ✓

C) are

D) have

Answer: B — was. 'Data' here is treated as uncountable/singular in this usage exam style: 'was'.

35. Choose the correct spelling:

A) Maintainance

B) Maintenance ✓

C) Maintenence

D) Maintainence

Answer: B — Maintenance. The correct spelling is 'maintenance'.

36. One word: 'A person who breaks into computers illegally' is a:

A) Programmer

B) Hacker (cracker) ✓

C) Analyst

D) Operator

Answer: B — Hacker (cracker). An illegal intruder into computer systems is a hacker/cracker.

37. The system crashed because ___ a power failure.

A) of ✓

B) off

C) for

D) with

Answer: A — of. 'Because of' is followed by a noun phrase.

38. Choose the synonym of 'Threat':

A) Promise

B) Danger ✓

C) Gift

D) Request

Answer: B — Danger. A threat is a danger or warning of harm.

39. She asked me ___ I had changed the password.

A) that

B) if ✓

C) what

D) which only

Answer: B — if. 'If' introduces a yes/no reported question.

40. The idiom 'a blessing in disguise' refers to:

A) A hidden good ✓

B) A secret enemy

C) A religious dress

D) A masked man

Answer: A — A hidden good. It means a seeming misfortune that turns out beneficial.

41. Neither the manager nor the assistants ___ aware of the breach.

A) was

B) were ✓

C) is

D) has

Answer: B — were. The verb agrees with the nearer subject (assistants — plural).

42. Choose the antonym of 'Vulnerable':

A) Weak

B) Open

C) Protected ✓

D) Easy

Answer: C — Protected. Vulnerable means open to attack; protected is the opposite.

43. Please do not share your password ___ anyone.

A) to

B) with ✓

C) for

D) by

Answer: B — with. We 'share something with someone'.

44. One word: 'Software that is free to use for a trial period' is:

A) Freeware

B) Shareware ✓

C) Firmware

D) Malware

Answer: B — Shareware. Shareware is distributed for trial before purchase.

45. The report must be submitted ___ Monday.

A) until

B) by ✓

C) to

D) till of

Answer: B — by. 'By' sets a deadline.

General Knowledge

46. The world's first programmable computer concept was designed by:

A) Alan Turing

B) Charles Babbage ✓

C) John von Neumann

D) Tim Berners-Lee

Answer: B — Charles Babbage. Babbage designed the Analytical Engine in the 1830s.

47. The World Wide Web was invented by:

A) Bill Gates

B) Tim Berners-Lee ✓

C) Steve Jobs

D) Vint Cerf

Answer: B — Tim Berners-Lee. Tim Berners-Lee invented the Web at CERN in 1989.

48. Which country launched the first artificial satellite, Sputnik 1?

A) USA

B) USSR (Russia) ✓

C) China

D) UK

Answer: B — USSR (Russia). The Soviet Union launched Sputnik 1 in 1957.

49. The largest social media company by users owns which platforms group?

A) Meta (Facebook, Instagram, WhatsApp) ✓

B) Alphabet only

C) ByteDance only

D) Microsoft only

Answer: A — Meta (Facebook, Instagram, WhatsApp). Meta owns Facebook, Instagram and WhatsApp.

50. The headquarters of the International Telecommunication Union (ITU) is in:

A) Geneva ✓

B) New York

C) Paris

D) Vienna

Answer: A — Geneva. ITU is headquartered in Geneva, Switzerland.

51. Pakistan Telecommunication Authority (PTA) regulates:

A) Banks

B) Telecom sector ✓

C) Railways

D) Airlines

Answer: B — Telecom sector. PTA regulates telecommunications in Pakistan.

52. The first electronic digital computer (general-purpose) is widely considered to be:

A) ENIAC ✓

B) UNIVAC

C) Abacus

D) Difference Engine

Answer: A — ENIAC. ENIAC (1945) was the first general-purpose electronic digital computer.

53. Which is the most widely used programming language for web interactivity?

A) Java

B) JavaScript ✓

C) C

D) Ruby

Answer: B — JavaScript. JavaScript runs in every web browser.

54. The '@' symbol in email addresses was popularised by:

A) Ray Tomlinson ✓

B) Tim Berners-Lee

C) Steve Wozniak

D) Linus Torvalds

Answer: A — Ray Tomlinson. Ray Tomlinson used @ for network email in 1971.

55. Pakistan's country domain extension is:

A) .pak

B) .pk ✓

C) .com.pk only

D) .gov

Answer: B — .pk. Pakistan's ccTLD is .pk.

56. The first mobile phone call was made in:

A) 1973 ✓

B) 1983

C) 1993

D) 2003

Answer: A — 1973. Martin Cooper made the first handheld mobile call in 1973.

57. Which organisation manages global IP address allocation at the top level?

A) ICANN ✓

B) PTA

C) ITU only

D) W3C

Answer: A — ICANN. ICANN coordinates the global DNS and IP addressing system.

58. The term 'computer bug' became famous after a real moth was found in which early computer?

A) ENIAC

B) Harvard Mark II ✓

C) UNIVAC

D) Colossus

Answer: B — Harvard Mark II. A moth was found in the Harvard Mark II in 1947.

59. The first search engine of the early web era was:

A) Google

B) Archie ✓

C) Yahoo

D) Bing

Answer: B — Archie. Archie (1990) is considered the first internet search engine.

60. Pakistan's first supercomputer was installed at:

A) NUST ✓

B) Quaid-i-Azam University

C) COMSATS

D) PIEAS

Answer: A — NUST. Pakistan's first supercomputer was installed at NUST, Islamabad.

Pakistan Studies

61. Who was elected President of Pakistan's first Constituent Assembly?

A) Liaquat Ali Khan

B) Quaid-e-Azam Muhammad Ali Jinnah ✓

C) Jogendra Nath Mandal

D) Khawaja Nazimuddin

Answer: B — Quaid-e-Azam Muhammad Ali Jinnah. Jinnah was elected President of the Constituent Assembly on 11 August 1947 (Mandal had chaired only the opening sitting as temporary president).

62. The Objectives Resolution (1949) was presented by:

A) Quaid-e-Azam

B) Liaquat Ali Khan ✓

C) Sardar Abdur Rab Nishtar

D) Zafarullah Khan

Answer: B — Liaquat Ali Khan. PM Liaquat Ali Khan presented the Objectives Resolution.

63. Pakistan joined the United Nations on:

A) 30 September 1947 ✓

B) 14 August 1947

C) 1 January 1948

D) 23 March 1948

Answer: A — 30 September 1947. Pakistan became a UN member on 30 September 1947.

64. The first recipient of Nishan-e-Haider was:

A) Major Muhammad Akram

B) Captain Muhammad Sarwar ✓

C) Major Tufail Muhammad

D) Sowar Muhammad Hussain

Answer: B — Captain Muhammad Sarwar. Captain Muhammad Sarwar Shaheed was the first Nishan-e-Haider recipient (1948 war).

65. Pakistan's national song 'Sohni Dharti' was written by:

A) Masroor Anwar ✓

B) Sahir Ludhianvi

C) Nisar Bazmi

D) Jamiluddin Aali

Answer: A — Masroor Anwar. Masroor Anwar wrote 'Sohni Dharti Allah Rakhay'.

66. The total number of seats in the National Assembly of Pakistan is:

A) 336 ✓

B) 342

C) 350

D) 326

Answer: A — 336. After the 25th Amendment/FATA merger, the National Assembly has 336 seats.

67. Pakistan's first ever Cricket World Cup win was in:

A) 1987

B) 1992 ✓

C) 1996

D) 1999

Answer: B — 1992. Pakistan won the Cricket World Cup in 1992 under Imran Khan.

68. The Chauburji monument is located in:

A) Multan

B) Lahore ✓

C) Karachi

D) Faisalabad

Answer: B — Lahore. Chauburji is a Mughal-era gateway in Lahore.

69. The first bank established in Pakistan after independence (central bank) started working on:

A) 1 July 1948 ✓

B) 14 August 1947

C) 1 January 1949

D) 23 March 1948

Answer: A — 1 July 1948. The State Bank of Pakistan began operations on 1 July 1948.

70. Pakistan Standard Time is how many hours ahead of GMT?

A) 4

B) 5 ✓

C) 6

D) 5.5

Answer: B — 5. Pakistan Standard Time is UTC+5.

Islamiyat

71. The Holy Quran was revealed over a period of about:

A) 13 years

B) 20 years

C) 23 years ✓

D) 25 years

Answer: C — 23 years. Revelation spanned about 23 years (13 in Makkah, 10 in Madinah).

72. The Surah without Bismillah at its start is:

A) Al-Anfal

B) At-Tawbah ✓

C) Yunus

D) Hud

Answer: B — At-Tawbah. Surah At-Tawbah begins without Bismillah.

73. The first martyr of Islam (male) among the early companions at Badr... the first Shaheed of Badr was:

A) Hazrat Hamza (RA)

B) Hazrat Mahja (RA) ✓

C) Hazrat Mus'ab (RA)

D) Hazrat Ubaydah (RA) only

Answer: B — Hazrat Mahja (RA). Hazrat Mahja (RA), a freed slave, is recorded as the first martyr at Badr.

74. The number of Surahs starting with 'Muqatta'at' (disjointed letters) is:

A) 26

B) 29 ✓

C) 31

D) 14

Answer: B — 29. 29 Surahs begin with Huroof-e-Muqatta'at.

75. Hazrat Ali (RA) was martyred in the city of:

A) Madinah

B) Kufa ✓

C) Damascus

D) Basra

Answer: B — Kufa. Hazrat Ali (RA) was martyred in Kufa (40 AH).

76. The Battle of Khandaq is also known as the Battle of:

A) Trench / Ahzab ✓

B) Badr

C) Moota

D) Qadsia

Answer: A — Trench / Ahzab. Khandaq means 'trench'; it is also called Ghazwa-e-Ahzab.

77. The first written constitution of Madinah is known as:

A) Misaq-e-Madinah ✓

B) Sulh Hudaybiyyah

C) Khutba Hajjatul Wida

D) Bait-e-Rizwan

Answer: A — Misaq-e-Madinah. The Charter of Madinah (Misaq-e-Madinah) was the first written constitution.

78. How many times has Hajj been declared obligatory in a Muslim's life?

A) Every year

B) Once in a lifetime (if able) ✓

C) Twice

D) Five times

Answer: B — Once in a lifetime (if able). Hajj is obligatory once in a lifetime for those who are able.

79. The Quran mentions the name of which companion by name?

A) Hazrat Abu Bakr (RA)

B) Hazrat Zaid bin Haritha (RA) ✓

C) Hazrat Umar (RA)

D) Hazrat Bilal (RA)

Answer: B — Hazrat Zaid bin Haritha (RA). Zaid bin Haritha (RA) is named in Surah Al-Ahzab (33:37).

80. The collection 'Muwatta' was compiled by:

A) Imam Malik ✓

B) Imam Abu Hanifa

C) Imam Shafi'i

D) Imam Ahmad

Answer: A — Imam Malik. Imam Malik ibn Anas compiled Al-Muwatta.

Basic Mathematics

81. What is 20% of 450?

A) 80

B) 85

C) 90 ✓

D) 100

Answer: C — 90. 20% of 450 = 0.20 × 450 = 90.

82. If a = 5 and b = 3, then $a^2 - b^2$ = ?

A) 16 ✓

B) 8

C) 25

D) 34

Answer: A — 16. 25 – 9 = 16.

83. The next number in the series 3, 9, 27, 81 is:

A) 162

B) 243 ✓

C) 108

D) 200

Answer: B — 243. Multiply by 3 each time; 81 × 3 = 243.

84. A data plan costs Rs 1,200 for 6 months. The monthly cost is:

A) Rs 150

B) Rs 180

C) Rs 200 ✓

D) Rs 240

Answer: C — Rs 200. 1200 ÷ 6 = Rs 200 per month.

85. What is the value of 2■?

A) 16

B) 32 ✓

C) 64

D) 10

Answer: B — 32. 2■ = 32.

86. The sum of 0.5 + 0.25 + 0.25 is:

A) 0.9

B) 1.0 ✓

C) 1.1

D) 1.25

Answer: B — 1.0. 0.5 + 0.25 + 0.25 = 1.0.

87. If 4 machines fill 400 bottles in a day, 1 machine fills:

A) 80

B) 90

C) 100 ✓

D) 120

Answer: C — 100. 400 ÷ 4 = 100 bottles.

88. The decimal 0.2 as a fraction in lowest terms is:

A) 1/2

B) 1/4

C) 1/5 ✓

D) 2/5

Answer: C — 1/5. 0.2 = 2/10 = 1/5.

89. A triangle with all sides equal is called:

A) Scalene

B) Isosceles

C) Equilateral ✓

D) Right-angled

Answer: C — Equilateral. An equilateral triangle has three equal sides.

90. What is 15 × 15 – 25?

A) 175

B) 200 ✓

C) 190

D) 225

Answer: B — 200. 225 – 25 = 200.

Everyday Science

91. The unit used to measure the speed of a computer processor is:

A) Hertz (GHz) ✓

B) Byte

C) Pixel

D) Volt

Answer: A — Hertz (GHz). Processor clock speed is measured in hertz, usually GHz.

92. Which gas is filled in electric bulbs (commonly)?

A) Oxygen

B) Argon ✓

C) Carbon dioxide

D) Hydrogen

Answer: B — Argon. Argon (an inert gas) is used in incandescent bulbs.

93. The memory of a computer is measured in:

A) Meters

B) Bytes ✓

C) Seconds

D) Watts

Answer: B — Bytes. Computer memory/storage is measured in bytes (KB, MB, GB).

94. Sunlight is a natural source of which vitamin?

A) Vitamin B12

B) Vitamin D ✓

C) Vitamin A only

D) Vitamin K

Answer: B — Vitamin D. Sunlight enables the skin to make Vitamin D.

95. The device used to measure earthquakes' intensity is:

A) Seismograph ✓

B) Barometer

C) Richter scale only (no device)

D) Thermometer

Answer: A — Seismograph. A seismograph records ground motion; the Richter scale rates magnitude.

Current Affairs & Cyber Law (Stable Facts)

96. NCCIA is Pakistan's agency dedicated to investigating:

A) Tax fraud

B) Cybercrime ✓

C) Smuggling of goods

D) Traffic offences

Answer: B — Cybercrime. The National Cyber Crime Investigation Agency investigates cybercrime in Pakistan.

97. NCCIA was created by upgrading/replacing the cybercrime role previously held by:

A) PTA

B) FIA Cybercrime Wing ✓

C) NADRA

D) PEMRA

Answer: B — FIA Cybercrime Wing. NCCIA took over cybercrime investigation from the FIA Cybercrime Wing (2024–25 transition).

98. The PECA law, dealing with electronic crimes in Pakistan, was originally passed in:

A) 2016 ✓

B) 2018

C) 2020

D) 2010

Answer: A — 2016. The Prevention of Electronic Crimes Act was passed in 2016.

99. Online harassment complaints by citizens in Pakistan can be filed with:

A) NCCIA ✓

B) Local post office

C) WAPDA

D) Railways

Answer: A — NCCIA. Citizens report cybercrime to the NCCIA.

100. Pakistan's national CERT deals with:

A) Currency printing

B) Cyber security incidents and response ✓

C) Census

D) Customs

Answer: B — Cyber security incidents and response. The national CERT (PkCERT) coordinates cyber incident response.